Newton's third law

LEARNING TARGET · R8 I can identify 3rd-law force pairs and explain why they never cancel.

A · KEY VOCABULARY

Interaction pair	When A pushes B, B pushes A at the same time.
Equal and opposite	Same size, same type of force, opposite directions.
Different receivers	One force acts on A, its partner acts on B — so they never cancel.
Effects differ	Equal forces, different masses → different accelerations.

B · KEY RULE

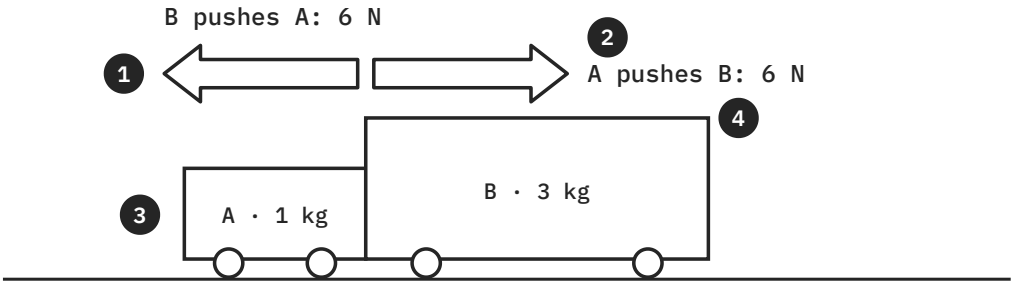
$$F_{A \text{ on } B} = -F_{B \text{ on } A}$$

swap the two object names

`force` newtons (N)
`–` opposite direction

C · COLOR IT · ONE INTERACTION, TWO FORCES

Each arrow belongs to a different cart.



- ☐ **1 · ORANGE** Force on A
- ☐ **2 · ORANGE** Force on B
- ☐ **3 · PENCIL** Cart A: label it
- ☐ **4 · PENCIL** Cart B: label it

D · WORKED EXAMPLE

Earth pulls an apple downward. Name the third-law partner force and the object that receives it.

GIVEN	EQUATION	WORK	ANSWER
Earth → apple gravity, down	swap the object names	apple → Earth gravity, up	**Apple pulls Earth up** Earth receives it

E · THE 2026-XJ THREAD

DAY 14 OF 22 · NEW TOOL: INTERACTION PAIRS

When Earth pulls on XJ, XJ pulls on Earth just as hard. The forces are equal, but XJ's far smaller mass means its motion is the one that changes noticeably.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else